

The Universe: An Overview

A Comprehensive Guide to Cosmic Knowledge

What is the Universe?

The Universe is all of space, time, matter, and energy that exists. It encompasses everything from subatomic particles to vast galaxy superclusters. Current scientific consensus holds that the Universe began approximately 13.8 billion years ago in an event known as the Big Bang — a rapid expansion from an extremely hot and dense initial state.

Scale and Structure

The observable Universe spans roughly 93 billion light-years in diameter. Within this vast expanse, matter is organized hierarchically: planets orbit stars, stars form galaxies, galaxies cluster into groups and superclusters, and superclusters form the cosmic web — filaments of dark matter and gas threading through vast cosmic voids.

Galaxies

There are an estimated 2 trillion galaxies in the observable Universe. Our own galaxy, the Milky Way, contains 100-400 billion stars and spans about 100,000 light-years. Galaxies come in various shapes: spiral (like the Milky Way), elliptical, and irregular. Most large galaxies harbor a supermassive black hole at their center.

Stars and Stellar Life Cycles

Stars are massive balls of plasma held together by gravity and powered by nuclear fusion. They form from collapsing clouds of gas and dust called nebulae. A star's life cycle depends on its mass: low-mass stars like our Sun become red giants and then white dwarfs, while massive stars explode as supernovae, sometimes leaving behind neutron stars or black holes.

Planets and Solar Systems

Planets form from the disk of gas and dust surrounding young stars (protoplanetary disks). Our Solar System contains 8 planets, dozens of moons, and billions of smaller bodies.

Astronomers have discovered over 5,500 exoplanets orbiting other stars, with thousands more candidates awaiting confirmation.

Dark Matter and Dark Energy

Ordinary matter makes up only about 5% of the Universe. Dark matter ($\approx 27\%$) is an invisible substance that exerts gravity but does not interact with light. Dark energy ($\approx 68\%$) is a mysterious force driving the accelerating expansion of the Universe. Neither dark matter nor dark energy has been directly detected; their existence is inferred from their gravitational and cosmological effects.

The Fate of the Universe

Several hypotheses describe the ultimate fate of the Universe. The most widely accepted is the 'Big Freeze' or Heat Death — as the Universe continues expanding, stars will eventually burn out, black holes will evaporate via Hawking radiation, and entropy will approach its maximum. Other theories include the Big Rip (dark energy tears apart matter) and the Big Crunch (gravity eventually reverses expansion).

Key Facts at a Glance

Property	Value
Age of Universe	~13.8 billion years
Observable Diameter	~93 billion light-years
Number of Galaxies	~2 trillion
Stars in Milky Way	100–400 billion
Composition (ordinary matter)	~5%
Dark Matter	~27%
Dark Energy	~68%
Expansion Rate (Hubble Constant)	~70 km/s/Mpc